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Dhirubhai Ambani Institute of Information and Communication Technology Gandhinagar

IT 607 Introduction to Database Management
InSem Exam-1, Autumn 2023-24

Closed Book Exam. Duration -1 Hour. 8 September 2023. Weightage 19%

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- This question paper has Three Sections. Write your answers strictly in the space provided.

Section A $4+2+0=6$

Questions to be answered strictly in the space provided. (4+3+3= 10 marks)

- Identify all 'PERSON' type entities you expect to find in the following well known databases. List all the important attributes and constraints for each 'Person' type entity which is part of each of the following databases:

a. Course Feedback Database of DA-IICT

Student (ID, first_name, last_name, department, email)

Instructor (ID, first_name, last_name, department, salary, specialization)

Teaching Assistant (ID, first_name, last_name, qualification, email)

b. Election Database of India

Leader (ID, name, party, region, contact_number, past_qualification)

Voters (ID, name, state, contact_number, eligibility, gender) region

Workers (ID, name, party, region, contact_number)

c. Resource Center Database of DA-IICT

Staff (ID, first_name, last_name, contact_number, section)

Borrower_details (ID, issue_date, name, contact_number, type) overdue

d. Hospital Database

Patients

entity sets:

Patients (patient_id, first_name, last_name, contact_number, email) Disease, Medical History

Doctors (doctor_id, first_name, last_name, specialization, email)

Staff (staff_id, first_name, last_name, department, shift)

2. Define and explain the following terms using concrete example (constructed by you):
 Note: Do not use examples cited by me in the class.

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a. **Participation Constraints** - There are two types of participation constraints: Total and partial. If all the entities of a particular entity set participate through the relationship with different entity sets then it is known as total participation. If all the entities do not participate in the relationship set then it is known as partial participation.

Ex. we have two entity sets: Customer and order.

Based on the diagram we can say that the order entity set has total participation and the customer table has partial participation.

customer	order
id 1	id 1
id 2	id 2
id 3	id 3
id 4	

b. **Candidate keys** - A candidate key helps to uniquely identify a tuple in the relation. For a candidate key no subset of it uniquely identifies the tuple.

e.g. For a doctors relation $\rightarrow$ doctors (doctor-id, first-name, last-name). Here we can see that ~~doctors~~ {doctor-id, first-name} can be considered as super key but not as candidate key because doctor-id alone can help uniquely identify the tuples. $\therefore$ candidate key = {doctor-id} Better say Can be given.

3. Develop a portion of the E-R Diagram (entities, relationships, attributes, keys, cardinalities etc) for the following paragraph which is part of the database of Academic Institute Management System:

An Instructor teaches one or more Courses. But every Course is taught by a single Instructor. Student studies one or more Courses. But some Courses may not have an enrollment at all. The Courses have Lectures. Some courses also have Tutorials. Every Course will have Coursework (Labs, Assignments) and Exams as Assessment Components.

entity sets:

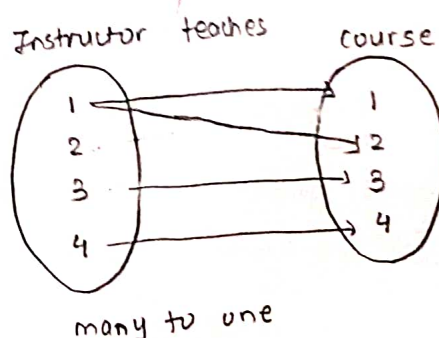
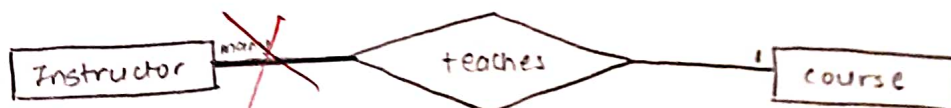
Instructor (ID, name, dept-name, salary)

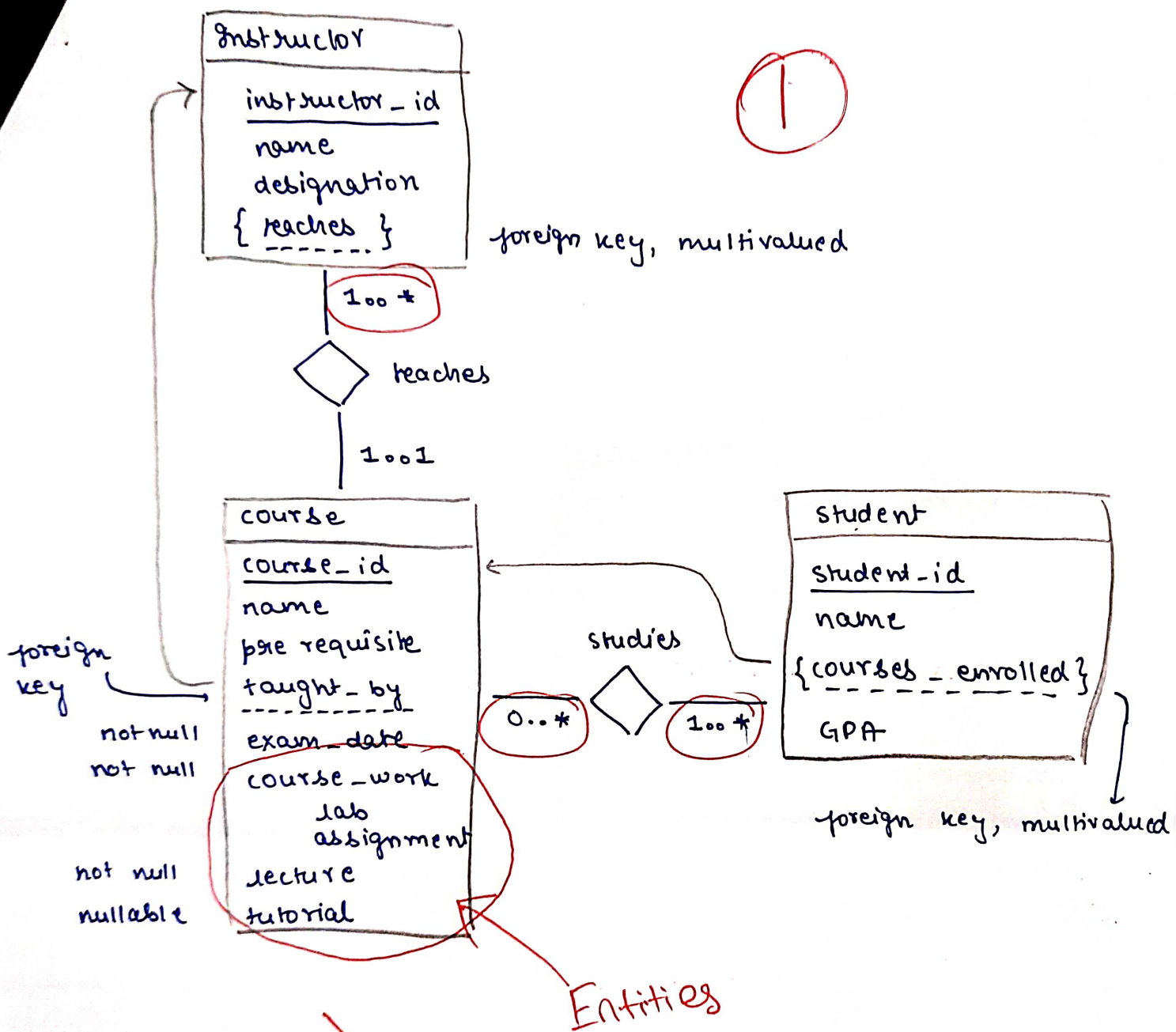
course (course-id, title, year, semester)

Student (ID, name, department, email)

Lecture (lecture-id, course-id, classroom, id)

foreign key
 ↓ references
 instructor





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Section B (3*2= 6 marks) NO PARTIAL MARKING

For the next 3 Questions, write down the output of the SQL queries

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1. Consider the table 'Grades' and the query given below:

Student	EHD_grade	Lab_grade
Akash	45	Null
Bhaskar	Null	90
Chanda	100	80

Write down the result of the query:

SELECT student

FROM Grades

WHERE (EHD_grade>Lab_grade AND Lab_grade>75 AND EHD_grade>90) OR (EHD_g

student
Akash
chanda

0

2. Suppose relation R (A, B) has tuples { (1,2), (1,2),(3,4) } and relation S (B, C) has tuples { (2,5),

Write down all tuples in the result of the SQL query:

SELECT *

FROM R NATURAL JOIN S

- 2 3. We have the following dealership data in the Dealership_table

<i>Dealership</i>	<i>Cars_sold</i>	<i>Profit</i>
Seva	100	50000
Aryan	NULL	NULL
Only BMW	NULL	40000
Ford Auto	50	25000

SELECT COUNT (Cars_sold) from Dealership_table returns:

2

2

2. Suppose relation R (A, B) has tuples { (1,2), (1,2),(3,4) } and relation S (B, C) has tuples { (2,5), (2,5), (4,6), (7,2) }.

Write down all tuples in the result of the SQL query:

SELECT *

FROM R NATURAL JOIN S

(1,2,5), (1,2,5), (1,2,5), (1,2,5), (3,4,6)

$$3 - 0.25 = 2.75$$

Section C (4*1= 4 marks) Overwriting or cancelling/scratching is considered as an incorrect answer.

Instructions

- For the next 4 questions, tick ✓ on the correct option for each.
- There is only one correct option for each question.
- You will get 1 for each correct answer, 0.25 for each incorrect answer. The wrong answer will fetch 0 marks.

- ✓ 1. Consider a relation $R(A, B)$ which may have duplicate tuples.

Query1:

`SELECT * FROM R;`

Query 2:

`(SELECT * FROM R)
INTERSECT
(SELECT * FROM R);`

- a. Query 1 and Query 2 always produce the same answer.
- b. The answer to Query 1 is always contained in the answer to Query 2.
- ✓ c. The answer to Query 2 is always contained in the answer to Query 1.
- d. Query 1 and Query 2 produce different answers.

- ✓ 2. Which of the following statements about E/R models is correct?

- a. Many-to-many relationships cannot be represented in E/R-diagrams
- ✓ b. Relationship sets may have attributes of their own.
- c. All many-to-one relationships are represented by a relationship between a weak and a non-weak entity set.
- d. None of the above

- ✗ 3. Consider the following query on the relation $R(A, B, C, D)$ Where A, B, C, D take integer values
- ```
SELECT [...]
FROM R
GROUP BY A, B
```

Which of the following can appear in the position marked as [...] in the select clause above

- I.  $MIN(C+D)$
- II.  $A, B$
- III.  $C, D$

- a. II only.
- b. I and II only.
- ✓ c. I, II, and III.
- d. None.

- ✓ 4. Following is one of the disadvantages of creating a 'view':

- ✓ a. It realizes only at the run time
- b. It wastes space and computational power
- c. Query processing with minimal optimization is not possible with views
- d. Security of data is compromised

3.

Consider the following query on the relation R (A, B, C, D) Where A,B,C, D take integer values

```
SELECT [...]
FROM R
GROUP BY A, B
```

Which of the following can appear in the position marked as [...] in the select clause above

- I.  $MIN(C+D)$
- II.  $A, B$
- III.  $C, D$

- a. II only.
- ☒ b. I and II only.
- c. I, II, and III.
- d. None.